USER GUIDE

Settlement Matching and Geocoding Tool

For IM Officers, GIS Specialists, and Data Assistants

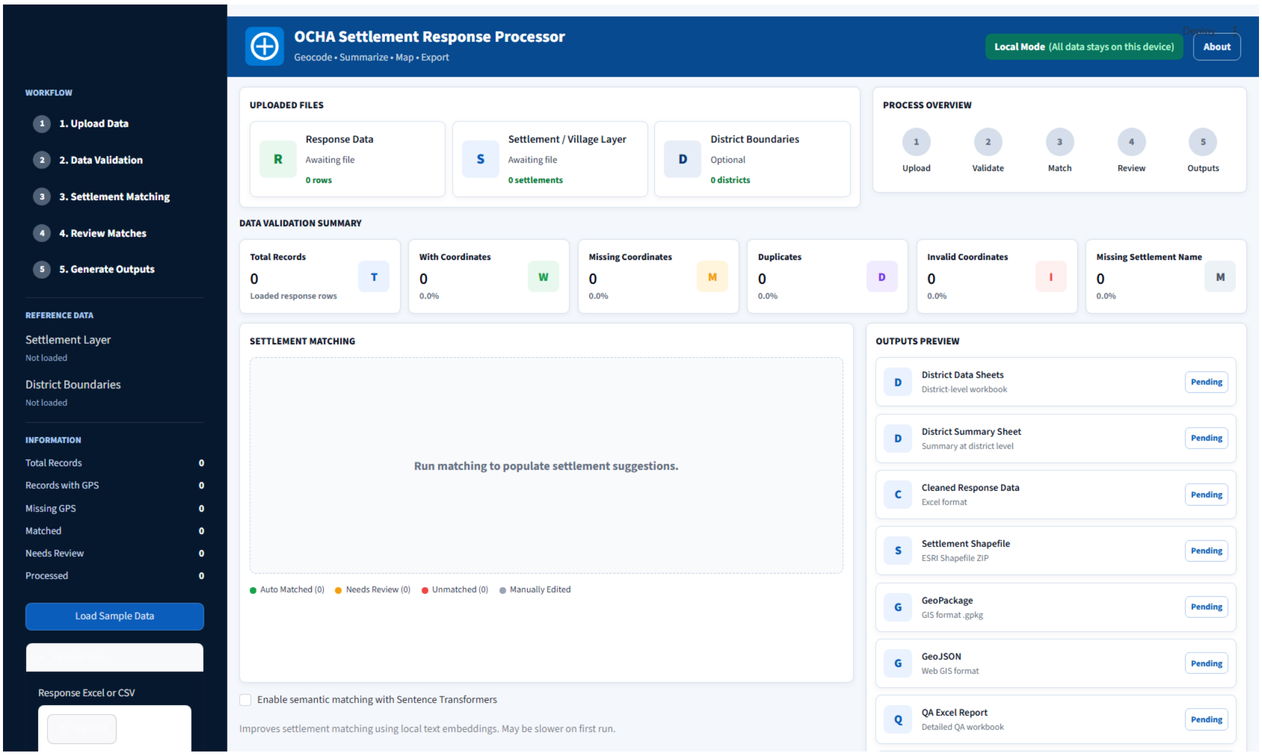
This guide explains how to use the app from a data-operator's point of view: what to prepare, what each screen shows, what the numbers and colors mean, and what to do when a record doesn't match automatically. It assumes no coding knowledge.

● Runs fully offline / Local Mode	No coding required	Streamlit desktop app

What this tool does

The processor takes a partner's response spreadsheet (settlements visited, beneficiaries reached, etc.), matches any records missing GPS coordinates against a settlement gazetteer, validates the data for common quality problems, and produces cleaned Excel workbooks, GIS files (Shapefile, GeoPackage, GeoJSON), and QA reports — ready to hand off or load into ArcGIS, QGIS, or Power BI.

> **Everything runs locally on the workstation.** No response data, gazetteer, or coordinates are sent to any external server. The header always shows a Local Mode badge as a reminder of this.



Landing screen, before any file is loaded

Before you start: what you need

Input	Required?	Must contain	Notes
Response data	**Required**	Settlement name, District	Lat/long optional — missing coordinates are what matching fills in. .csv, .xlsx, .xls.
Settlement gazetteer	**Required**	Settlement, District, Lat, Long	Region recommended. Also accepts a spatial file (.geojson, .gpkg, .zip shapefile).
District boundaries	**Optional**	—	Adds a boundary overlay to the map. .geojson, .json, .gpkg, .zip.

You don't have to name columns exactly — common humanitarian naming variants are recognized automatically:

- **Settlement** → settlement, village, site, location, town, …

- **District** → district, admin2, adm2_en, …
- **Region** → region, admin1, adm1_en, …
- **Latitude / Longitude** → lat/lon, y/x, gps_latitude/gps_longitude, …
- Optional fields like **partner**, **cluster/sector**, and **beneficiaries** are also detected and carried into outputs.

If a required column can't be found, Data Validation (Step 2) says so explicitly and names the missing field.

> **No files yet?** Click Load Sample Data in the sidebar for a ready-made example (12 response rows, 11 gazetteer settlements, 5 districts) and try the whole workflow before touching real data.

The five-step workflow

The sidebar tracks progress through five stages — a stage turns green once complete, and the current one is highlighted. The same steps appear as a horizontal tracker at the top of the main panel.

01 Upload Data	**02** Data Validation	**03** Settlement Matching	**04** Review Matches	**05** Generate Outputs

Step 1 — Upload Data

In the sidebar:

- **Load Sample Data** — instantly loads the bundled example dataset.
- **Upload files** (expandable) — three pickers: Response Excel/CSV, Settlement Gazetteer, District Boundary Layer (optional). Choose files, then click Load Uploaded Files.

Once loaded, the Uploaded Files panel confirms what was read: file name, response row count, gazetteer settlement count, boundary district count.

To start over completely, use **Restart Process** in the sidebar — it clears all loaded data, matches, and generated files.

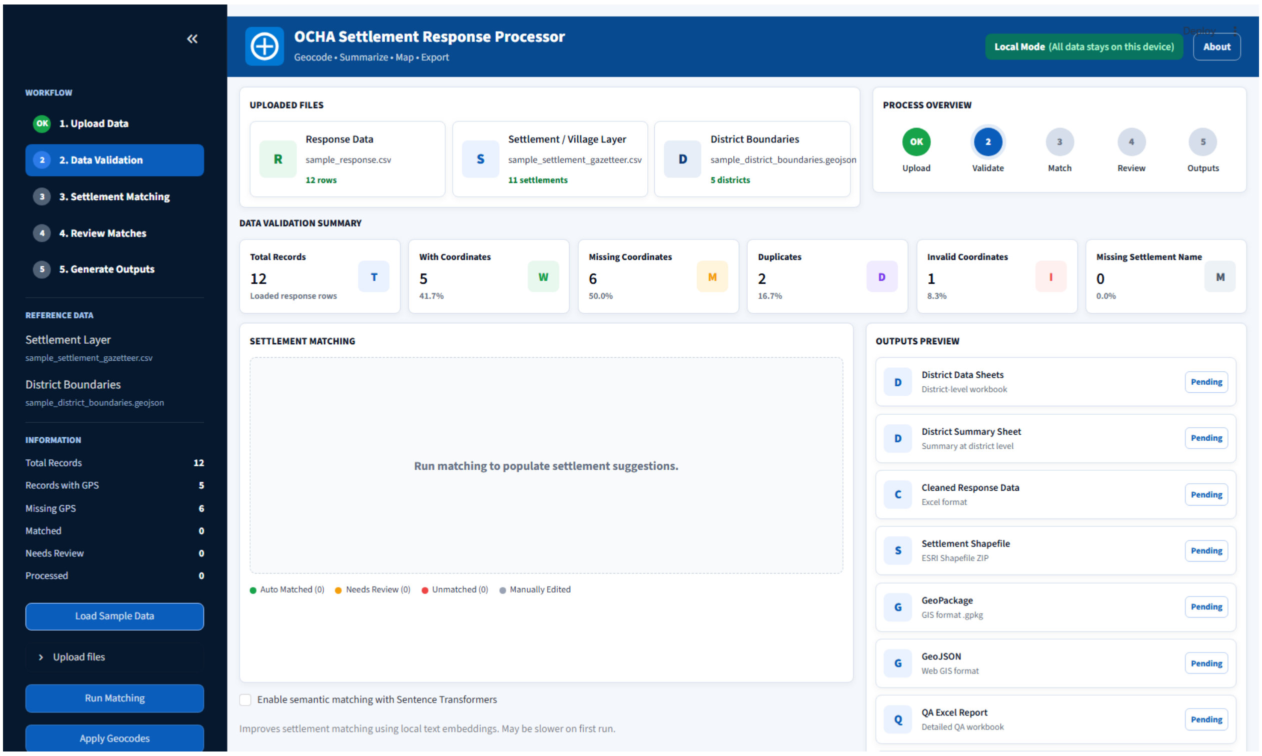
Step 2 — Data Validation

Six at-a-glance metrics appear before matching runs.

Metric	What it means
Total Records	Rows read from the response file.
With Coordinates	Already have valid lat/long — untouched by matching.
Missing Coordinates	No GPS value — these go to settlement matching.
Duplicates	Rows that look like repeats (same settlement, district, partner, cluster). Worth a manual check.
Invalid Coordinates	Lat/long outside valid range (lat –90 to 90, lon –180 to 180) — likely swapped or mistyped.
Missing Settlement Name	No settlement name at all — can't be auto-matched; fix at the source.

The app also checks for districts absent from your gazetteer, and region/district pairs that don't match its hierarchy — both flagged as issues with a red (blocking) or yellow (worth reviewing) severity.

> **Practical tip:** if Missing Settlement Name or Invalid Coordinates is non-zero, fix those in the source spreadsheet and re-upload — no automated step can recover a record with no name or an impossible coordinate.



Uploaded files and validation summary, right after loading data

Step 3 — Settlement Matching

Click **Run Settlement Matching** to attempt to geocode every response row missing coordinates.

How matching works, in order

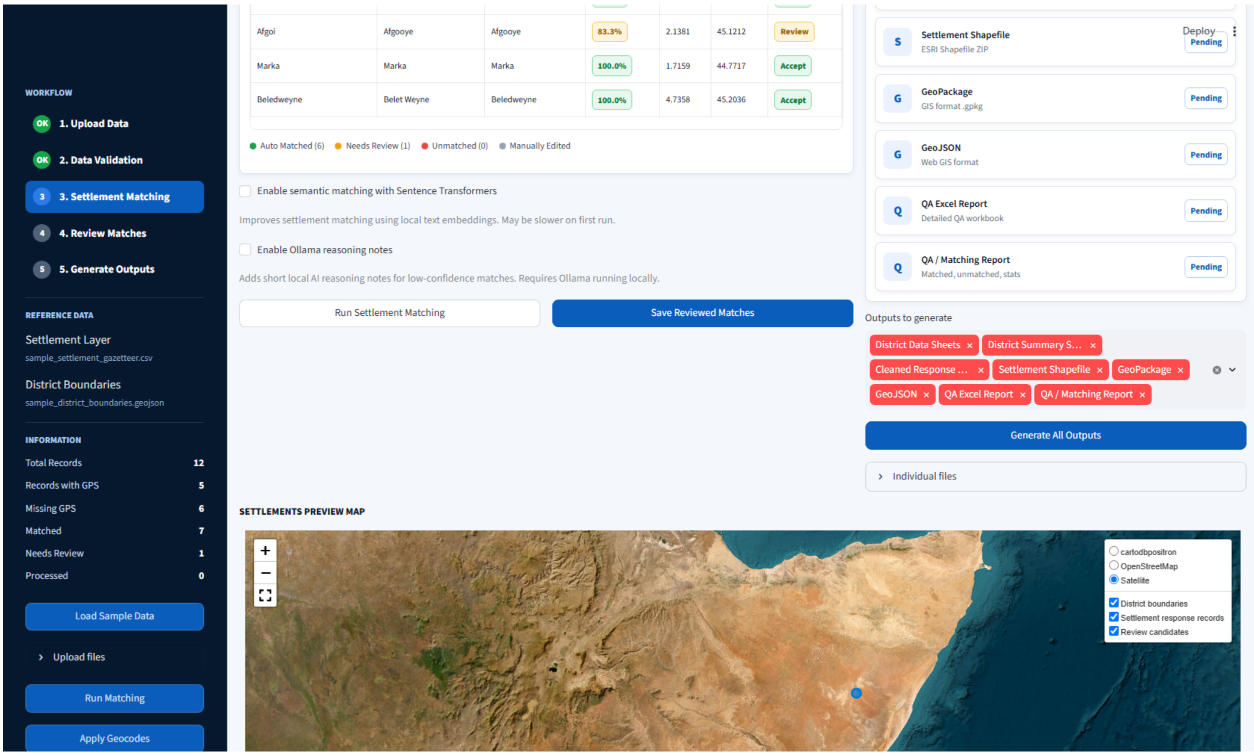
1. **Exact match** — normalized settlement name matches the gazetteer exactly.
2. **RapidFuzz fuzzy match** (always on) — scores the closest gazetteer candidates by text similarity, narrowed by district/region when available.
3. **Semantic matching** (optional toggle) — a local Sentence Transformers embedding model (all-MiniLM-L6-v2) catches names worded differently but meaning the same place. First run downloads the model once.
4. **Ollama reasoning notes** (optional toggle) — for “needs review” matches, asks a local Ollama LLM (qwen2.5) for a one-line plain-English reason. Requires Ollama running locally (ollama serve); if unreachable, the app warns and simply skips this step.

Both AI toggles are **off by default** and layer on top of RapidFuzz — never replace it. If a toggle is on but its dependency is missing, the app warns you and continues with RapidFuzz-only matching so a run never fails outright.

Confidence scoring and status

Component	Weight
Settlement name similarity	50%
District similarity	25%
Region similarity	15%
Administrative consistency	10%

Confidence	Status	What happens
90-100%	Auto Matched	Coordinates applied automatically.
75-89%	Needs Review	Flagged for manual follow-up — see Step 4.
0-74%	Unmatched	No confident candidate found.



Matching table with confidence scores, AI toggles, and outputs preview

Step 4 — Review Matches

“Needs Review” and “Unmatched” records deserve attention — the app deliberately never auto-applies anything below 90% confidence, so a human always has the final say on lower-confidence geocodes.

> **Good to know:** this build doesn't yet include an in-app grid for accepting or overriding individual low-confidence matches row-by-row. Resolve them one of these ways instead.

- **Fix the spelling at the source.** Most “needs review” cases are a spelling or transliteration difference. Correct it in the response file and re-run matching.
- **Turn on semantic matching** if you haven't — it often resolves near-miss spellings RapidFuzz alone scores lower.
- **Check the QA Excel Report's “Low Confidence” sheet** (Step 5) — every needs-review/unmatched record with its suggested candidate side by side.
- **Source the coordinate manually** and enter it into the response file, then re-upload — the most reliable fix for settlements genuinely missing from the gazetteer.

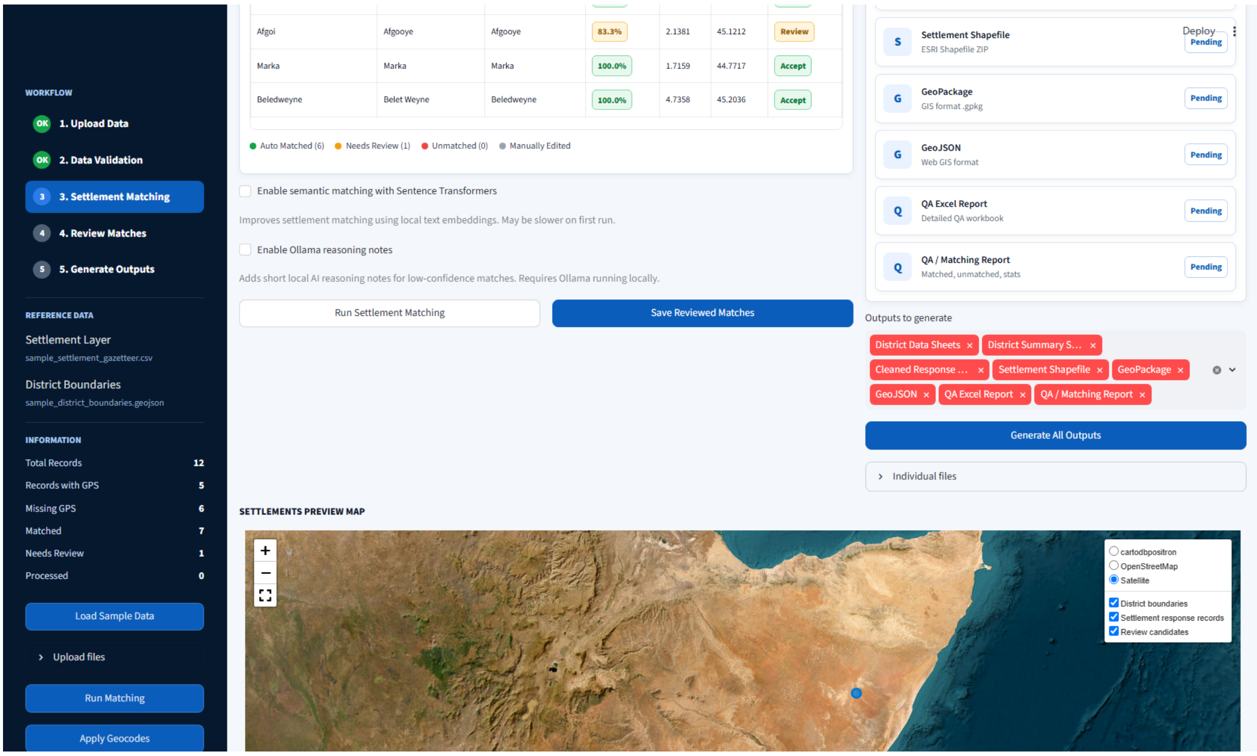
Once satisfied, click **Save Reviewed Matches** and then **Apply Geocodes** (sidebar) to write accepted coordinates back into the working dataset used for outputs and the map.

The Settlements Preview Map

A full-width, interactive view of every geocoded record sits below the matching and outputs panels.

- **Base layers** — switch between CartoDB Positron, OpenStreetMap, and Esri Satellite via the layer control (top right).
- **Overlays** — toggle district boundaries, settlement response records (clustered, colored by status), and review candidates independently.
- **Click any marker** for settlement name, district, match status, and confidence.
- **Zoom controls** (top left) plus normal scroll-wheel/drag, like any GIS viewer.
- **Fullscreen button** (top left) — expands the map to fill the browser window.
- **Mini-map** (bottom left) — shows your current viewport in the wider region.

The map updates automatically as you load data, run matching, and apply geocodes — no manual refresh needed.



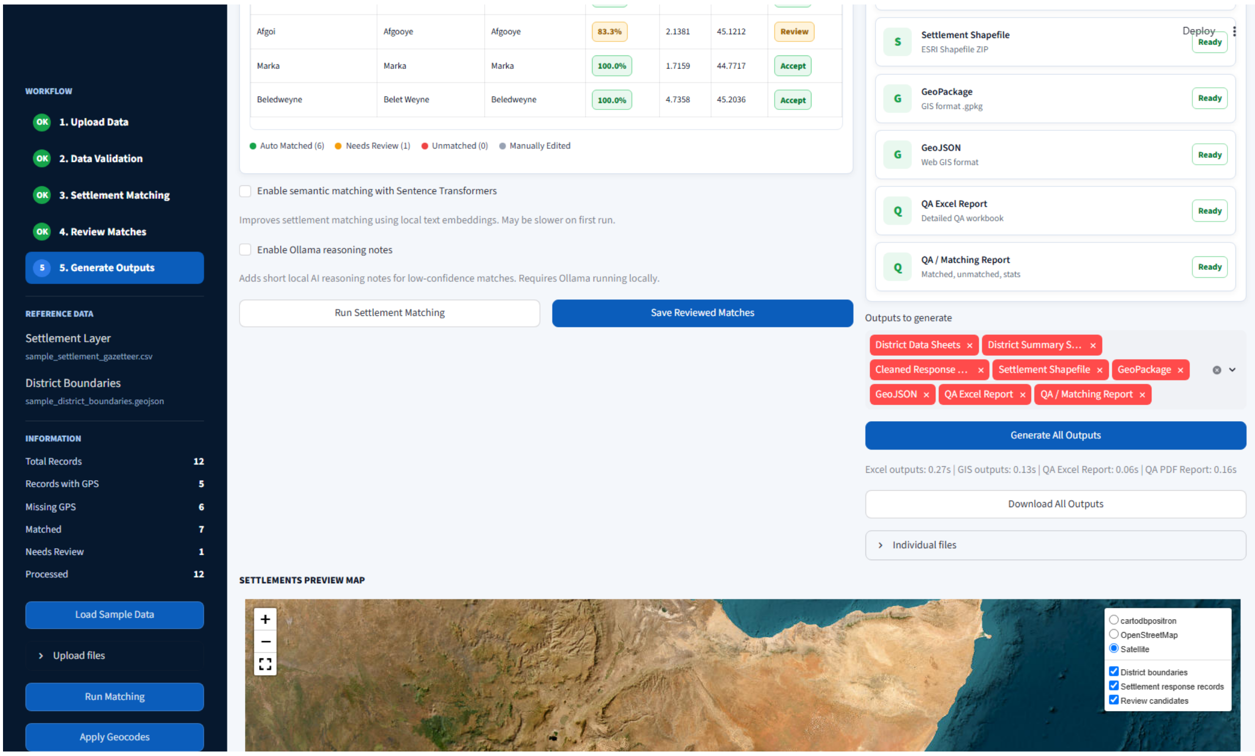
Full-width map, satellite base layer, layer control open

Step 5 — Generate Outputs

Use **Outputs to generate** to choose which files you need (all selected by default), then click **Generate All Outputs** (or **Generate Selected Outputs** if narrowed down).

Output	Format	Contents
District Data Sheets	Excel	One sheet per district with response records.
District Summary Sheet	Excel	Aggregated totals by district (and cluster, where available).
Cleaned Response Data	Excel	Full response dataset with applied geocodes and match metadata.
Settlement Shapefile	ESRI Shapefile (zipped)	Point layer for GIS software.
GeoPackage	.gpkg	Same data, modern GIS format.
GeoJSON	.geojson	Web-GIS-friendly point layer.
QA Excel Report	Excel	Readiness metrics, validation issues, full match table, “Low Confidence” sheet, district/cluster summaries.
QA / Matching Report	PDF	Shareable summary of matching results and data quality.

A processing log (ocha_processing_log.txt) and a combined **Download All Outputs** ZIP are generated automatically alongside your selected files. Each output card shows Ready once generated, or an error message if a stage failed.



Outputs generated and ready to download

Reading the sidebar at a glance

The sidebar's Information panel is a live dashboard of where you stand:

- **Total Records** — rows in the response file.
- **Records with GPS** — already had valid coordinates on upload.
- **Missing GPS** — still need matching.
- **Matched** — auto-matched + needs-review records combined.
- **Needs Review** — subset of Matched not yet at auto-accept confidence.
- **Processed** — rows in the final dataset (outputs/map) after geocodes are applied.

The Reference Data section above it reminds you which gazetteer and boundary files are currently loaded.

Troubleshooting / FAQ

“sentence-transformers is not installed” warning

Semantic matching was on but the package isn't available. The run still completes using RapidFuzz only. Ask whoever manages the installation to run pip install sentence-transformers, then try again.

“Ollama is not reachable at localhost:11434” warning

Ollama reasoning notes were on, but no local Ollama server is running. The run still completes without the extra notes. To enable: install Ollama, run ollama pull qwen2.5, then ollama serve, and re-run matching.

A settlement I know exists isn't matching at all

Check it's actually in the gazetteer (correct spelling, correct district). If genuinely missing, no amount of fuzzy or semantic matching will find it — add it to the gazetteer, or enter the coordinate manually.

The map shows fewer points than my total record count

The map only plots rows with a valid coordinate. Rows still Needs Review or Unmatched won't appear until resolved (Step 4) and geocodes applied.

I want to start over with a different dataset

Click Restart Process in the sidebar — clears loaded files, matches, and generated outputs without reloading the app.

Is any of my data leaving this computer?

No. Core matching (RapidFuzz, Sentence Transformers) runs locally, and Ollama, if used, is also a local process. The header's Local Mode badge is a permanent reminder of this.

Quick reference

Range	Status
90-100%	Auto Matched
75-89%	Needs Review
0-74%	Unmatched

Color	Meaning
Green	Auto-accepted / already geocoded
Amber	Needs review
Red	Unresolved / invalid
Grey	Manually edited / rejected

Dataset	Required fields
Response data	Settlement, District
Gazetteer	Settlement, District, Lat, Long